The Hygiene Trap

Sampling day sheet: paired first-draw samples

Repeats the design of Swanson and He (2024). Plates stay sealed and are never opened after they have grown. Anyone with a weak immune system should not handle them. Full method: replication protocol and plate protocol at https://github.com/aptitudetechnology/fridge-water.

On the day

☐ No use of the fridge dispenser or the cold kitchen tap for 8 hours or more. Write down when each was last used.
☐ Wash hands, gloves on. **Do not clean the spout first**: you want what a glass would get.
☐ Fridge first draw (F1) into a sterile bottle, without letting the bottle touch the spout.
☐ Tap first draw (T0) from the cold kitchen tap, without letting it run first.
☐ Chlorine test on both, at once (within a minute). Ask the water supplier: chlorine or chloramine? With chloramine, use total chlorine.
☐ Optional: flushed fridge sample (F2, after running 2 litres), nozzle swab (N), drip tray swab (D), blank (B).
☐ Plate within 2 hours, keep bottles cold. Incubate sealed, at room temperature only.

About this home

Home code (anonymous) _____ Region (state or country only) _____ Date _____

Fridge make and model (optional) _____ Filter type _____
Filter age in months _____

Supplier disinfectant: ☐ chlorine ☐ chloramine ☐ unknown ☐ private supply Hours since last use: fridge _____ tap _____

Readings

Sample	Chlorine (free or total)	Volume plated (mL)	Colonies, day 3	Day 5	Day 7	CFU per mL	Notes
T0 tap first draw							
F1 fridge first draw							
F2 fridge flushed (optional)							
N nozzle swab (optional)							
D drip tray swab (optional)							
B blank							

Record a tap plate with no colonies as "below 1 CFU per mL", not as zero. A blank that grew means your handling added bacteria: the result is unreliable. Our suggested rule, written before you look: the fridge matches the original if it is at least 100 CFU per mL, at least 10 times the tap (or the tap is below detection), and has less chlorine than the tap.

Share it: copy the rows into replication_results_template.csv and use the "Replication results" form at github.com/aptitudetechnology/fridge-water/issues. Region only: no address, no names, no photos of your home.

References

1. Swanson CS, He Q (2024). Linkage of in-refrigerator water dispensing systems to elevated exposure to microbial contamination at the microbiome scale. Am J Infect Control 52:857-859. PMID 38460731. doi:10.1016/j.ajic.2024.03.001

Source: The Hygiene Trap, print pages. https://fridge-water.ownhealth.bio/fridge-water#handouts (whole site: https://fridge-water.ownhealth.bio/). Printed November 6, 2026.
Every claim carries a label: established, plausible, untested, anecdote or real. This is a hypothesis, not a diagnosis: for persistent bad breath or symptoms, see a dentist and a doctor.
Text: CC BY 4.0. Please credit the Fridge water project, https://github.com/aptitudetechnology/fridge-water.